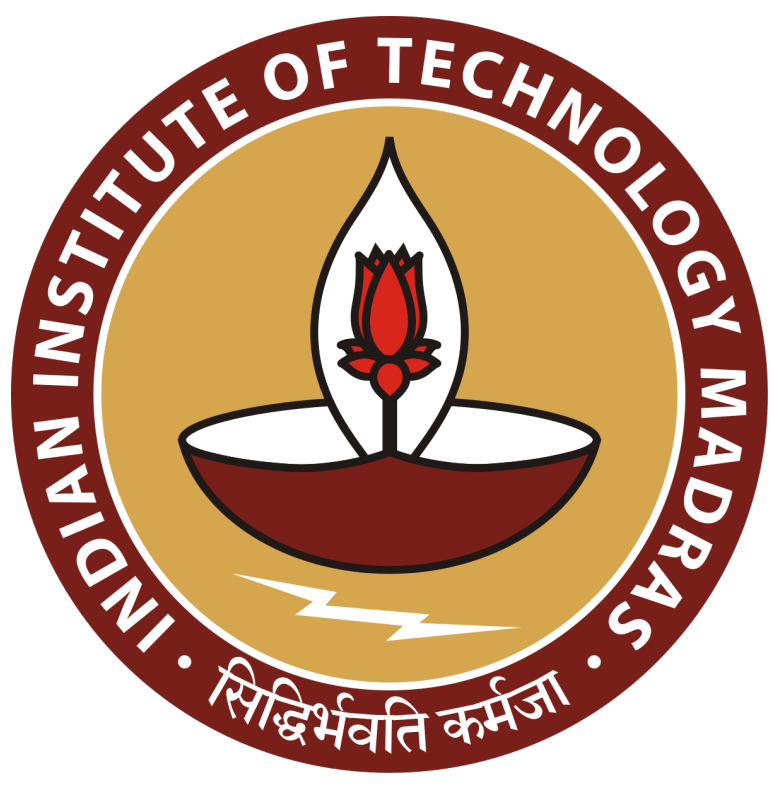




Digital Twin - Based Multi Echelon Inventory Optimization

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Introduction

Background and Motivation

Inventory management decisions play a central role in determining the efficiency, resilience, and cost-effectiveness of modern supply chains. Organizations operating multi-echelon inventory networks must make critical replenishment decisions under demand uncertainty, transportation limitations, and operational constraints, often without clear visibility into system-wide consequences. As a result, modifying inventory policies in real-world systems is inherently risky, since downstream impacts on service levels, costs, and stability are difficult to anticipate in advance.

While a wide range of analytical models exist for multi-echelon inventory systems, their assumptions often simplify demand dynamics, transportation constraints, or network structure. In practice, decision-makers frequently rely on simulation tools to evaluate inventory policies and operational scenarios before deployment. However, many such tools are proprietary, domain-specific, or tightly coupled to specific industrial settings, limiting their transparency and extensibility.

Key contribution

This work addresses this gap by developing a **digital twin-based simulator** for multi-echelon inventory systems with explicit modeling of stochastic demand, transportation capacity constraints, and cost components. The current phase focuses on building and validating the core simulation modules and their interactions to ensure realistic system behavior. The framework is designed to be **modular, extensible and open source**, serving as a foundation for future studies on inventory optimization including the impact of demand forecasting on policy selection, transportation policy evaluation and cost-service trade-offs.

Data

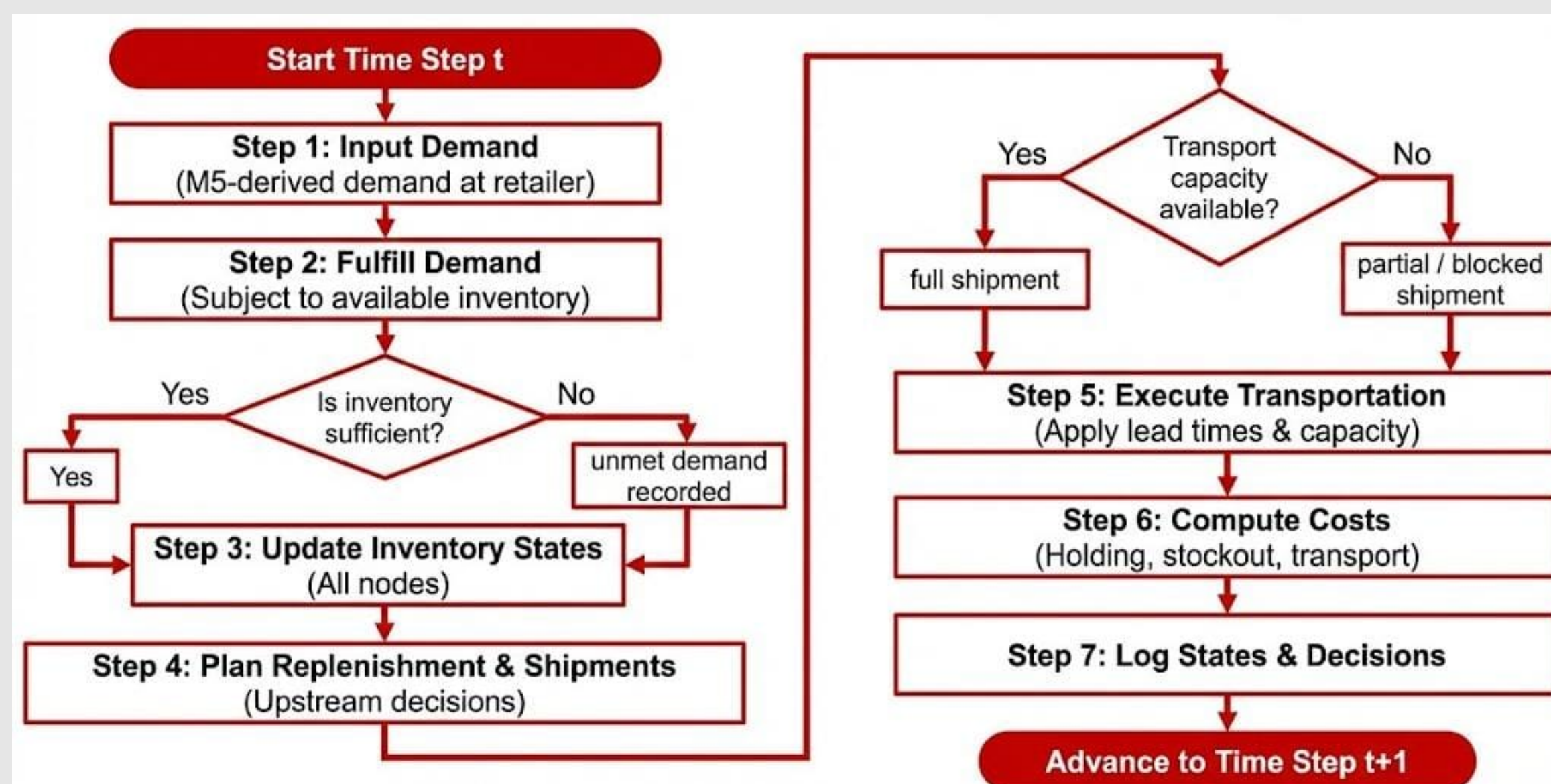
- **Demand Data:** Retail-level demand is provided either as generated time-series or as processed data derived from the M5 forecasting dataset (Walmart sales data).
- **Inventory Data:** Initial inventory levels, storage capacities, and replenishment parameters are defined for each echelon of the supply chain.
- **Transportation Data:** Transportation links are defined through route connectivity, capacity constraints, and lead times governing material flow between echelons.
- **Cost Parameters:** Holding costs, stockout penalties, and transportation costs are specified to evaluate cost-service trade-offs.

Non-demand parameters are defined to enable systematic testing and validation. The simulator is designed to directly take in real-world operational inputs without changes to the simulation logic.

System Architecture and Simulator Design

The digital twin represents a multi-echelon supply chain as a directed network of inventory-holding nodes (supplier, warehouse, retailer) connected by transportation links with finite capacity and lead times. Each node maintains a local inventory state, while transportation links explicitly govern feasible material flow between echelons.

Discrete-Time Simulation Flow of the Digital Twin



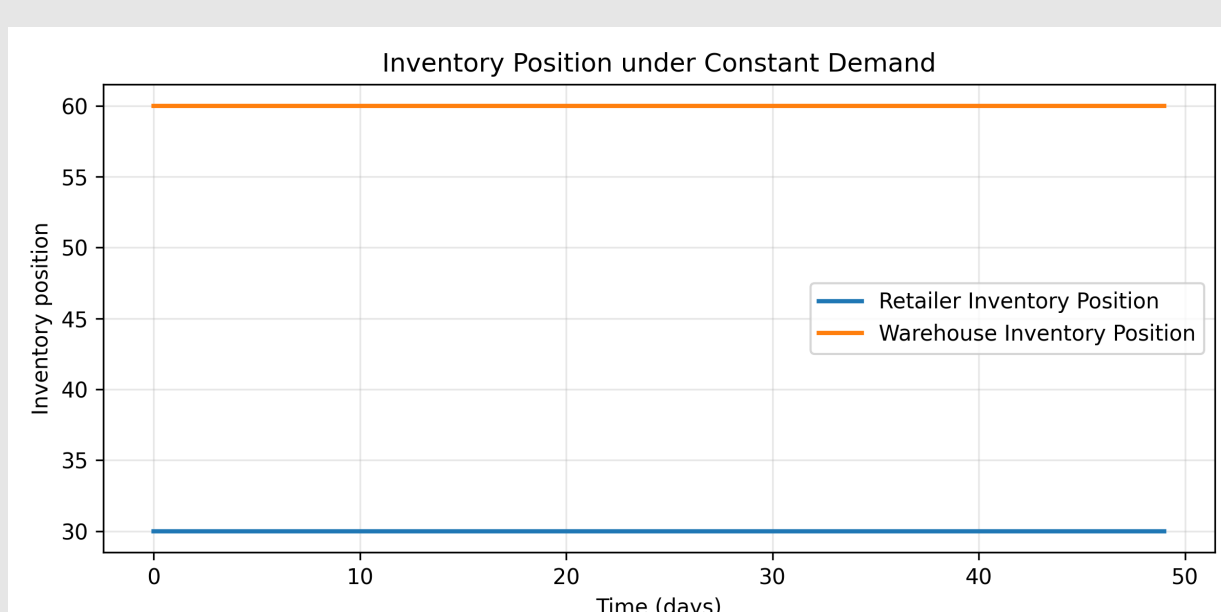
Simulator Features

- **Flexible Input Interface:** The simulator accepts structured inputs via CSV configuration files or as deterministic parameter definitions in json, enabling easy switching between experimental setups and real-world data sources.
- **Multi-Echelon and Multi-Route Modeling:** Arbitrary supply chain topologies are supported, including one-many, many-one and many-many connections between nodes and multiple transportation routes between the same pair of nodes with independent capacities and lead times.
- **Inventory Policy Support:** Each inventory-holding node applies a configurable replenishment policy. The current implementation supports base-stock and (s, S) - type policies, with a modular design for extending to additional policies.
- **SKU-Level Extensibility:** The simulator is designed to support both single-SKU experiments and extension to multiple SKUs without changes to the core execution logic.
- **Shipment Planning and Execution:** Replenishment decisions generate shipment requests that are executed subject to transportation feasibility, allowing partial or blocked shipments and realistic delay propagation - modifiable by changing the transportation policy.

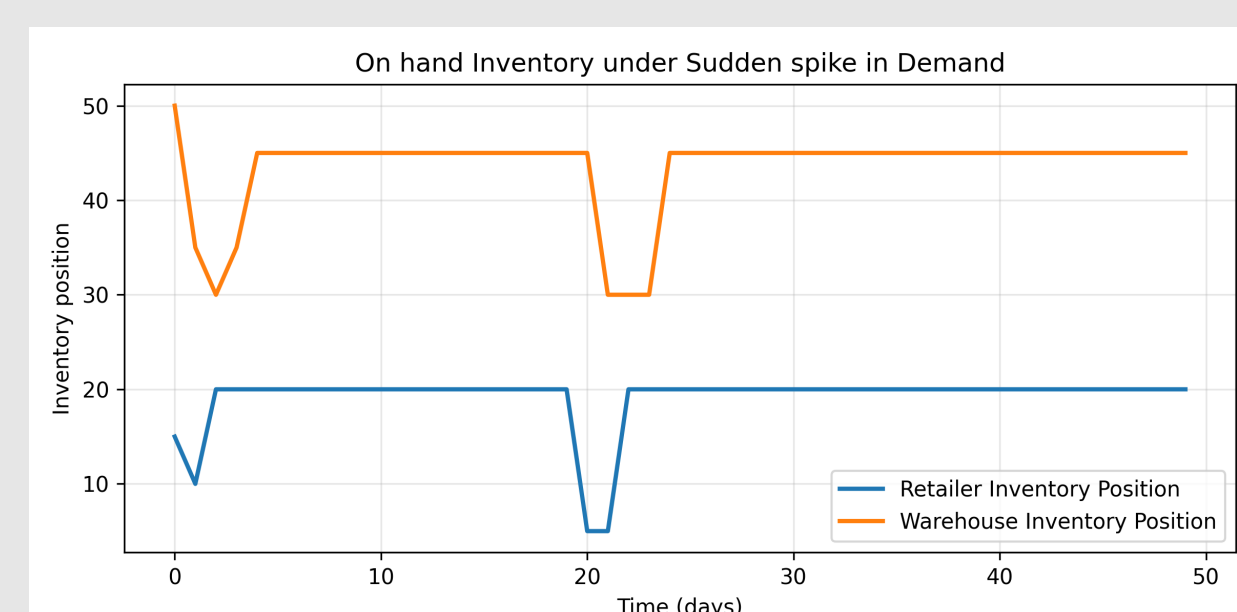
Results

Simulator Validation and Demand Shock Analysis

- **Constant-demand validation (Fig. a):** A simple three-echelon 1 Supplier - 1 Warehouse - 1 Retailer network is simulated with an infinite-supply source and base-stock policies. Retailer demand is fixed at 5 units/day with constant lead times (2 and 1 days), and inventory position is plotted.
- **Demand shock experiment (Fig. b):** Using the same network and policy, a one-day demand spike of 20 units is introduced at the retailer. On-hand inventory is plotted to observe the delayed upstream response caused by lead times and transportation constraints.



(a) Inventory Position under Constant Demand



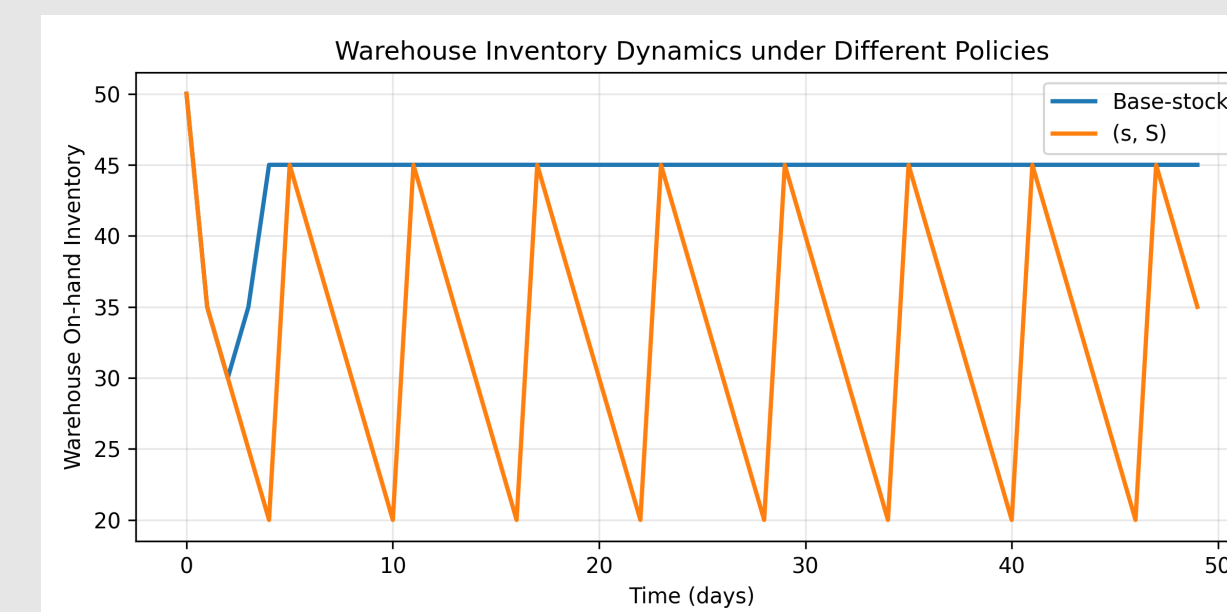
(b) On-hand Inventory under Demand Spike

Results

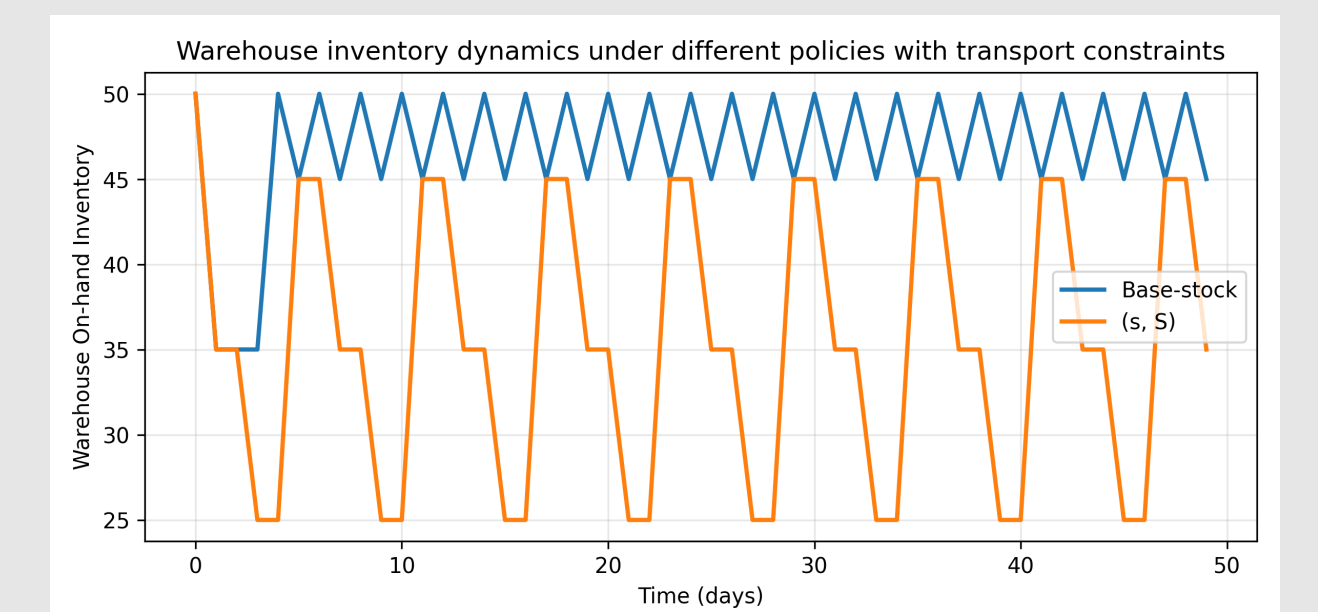
- Under constant demand, the inventory position at both the retailer and warehouse remains regulated at the base-stock level, validating the ordering and pipeline logic.
- When a demand spike is introduced, inventory position remains regulated due to policy response, while on-hand inventory shows an initial transient dip due to system warm-up, followed by an immediate retailer dip and a delayed warehouse response caused by lead times.

Policy and Transport Constraints Analysis

We compare **base-stock** and **(s, S)** policies on the **same three-echelon synthetic network** used in Fig. (a) and Fig. (b). In an (s, S) policy, replenishment is triggered only when inventory falls below a reorder point s , and orders raise inventory up to a target level S . Only warehouse policy and transportation capacity of a single link (warehouse to retailer) are varied.



(c) Warehouse inventory without transport delay



(d) Warehouse inventory with transport delay

- **Without transport constraints (Fig. c):** Base-stock stabilizes at an on-hand level of 45, corresponding to a target of 60 units minus 3 days of pipeline inventory (3×5) . The (s, S) policy oscillates between 45 and 20, as replenishment is triggered only when inventory falls below $s = 30$, causing deeper troughs before recovery.
- **With transport constraints (Fig. d):** Imposing a capacity limit on the warehouse-retailer link raises the effective on-hand inventory for both policies. Under base-stock, inventory no longer stabilizes at 45 but oscillates between 45 and 50 due to shipment batching at the capacity limit. Under the (s, S) policy, the minimum inventory increases from 20 to 25, as the capacity constraint prevents further drawdown once the shipment limit is reached.

Transport Capacity-Cost-Service Trade-offs

- Using a two-warehouse, three-retailer network (One warehouse connects to two retailers) with **M5 demand data**, we simulate and evaluate 5 scenarios by varying transportation capacities.
- The baseline transport capacities and cost parameters are assumed based on the scale of the M5 demand data to enable controlled experimentation

Table 1. Impact of Transport Capacity on Cost and Service Level (Base-stock Policy)

Capacity	Total Cost	Fill Rate (%)
0.25× Baseline	73,438	49.3
0.50× Baseline	102,416	84.4
1.00× Baseline	184,295	98.0
2.00× Baseline	346,500	99.2
4.00× Baseline	712,235	99.3

- The results show a sharp non-linear trade-off: increasing capacity rapidly improves fill rate from 49% to ~98% at baseline capacity, but further capacity increases primarily raise transport cost with marginal service improvement.

Conclusion

- This work presents a modular, open-source digital twin framework for simulating multi-echelon inventory systems. Controlled experiments reproduce key system behaviors such as pipeline delays, shipment batching, and policy-dependent inventory oscillations, validating the correctness of the core simulation logic.
- Policy-level comparisons show that transportation capacity significantly alters inventory dynamics even under identical demand and lead-time settings. Capacity constraints induce delayed replenishment and higher inventory variability, demonstrating the need to jointly model inventory and transportation decisions.
- Using real-world M5 demand data, a systematic capacity-scaling study highlights a clear cost-service trade-off. Fill rate improves as transport capacity increases up to a baseline threshold, beyond which additional capacity increases total costs with limited service-level gains.
- As a next phase, the framework will be extended toward more realistic decision-making settings, including multi-SKU interactions, adaptive and learning-based inventory policies, richer transportation networks, and demand-forecast integration. These extensions aim to support digital twin-driven policy evaluation, capacity planning, and optimization under realistic supply chain uncertainty.

References

- Katsov, I. *TensorHouse: Optimization framework* - <https://github.com/ikatsov/tensor-house>
- M5 Forecasting Dataset, Kaggle - <https://www.kaggle.com/c/m5-forecasting-accuracy>